

LINUX PROGRAMMING ASSIGNMENT - 05

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SECTION B
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ROLL NO. 21

1. **What is a shell in Linux OS? How many categories of shell is currently exists in Linux? Why bash shell is very popular in Linux distribution?(CO2)**

Answer - A shell is a command-line interface that allows users to

interact with the Linux system.

There are mainly two types of shells:-

- command-line shell
- graphical shell
- the bash shell is most popular because it is user-friendly, fast and supports scripting.

2. **What does ls-z command display?(CO1)**

Answer - ls-z shows the SELinux security context of files and directories, like their labels and permissions.

3. **Write a command to list all hidden files in the current directory.(CO1)**

Answer -

ls -a

4. **Explain the difference between hard links and soft links in Linux.**

(CO1)

Answer -

Hard link - points directly to the files data on disk. Even if the original file is deleted, the data remains.

Soft link - points to the file name, not data. If the original file is deleted, the link breaks.

5. A file has permissions `-rwxr-x-x`. Explain who can read, write and execute it. (CO1)

Answer -

Owner - read, write, execute.

Group - read and execute.

Others - execute only.

6. Write the command to change the group ownership of a file `data.txt` to group `staff`. (CO1)

Answer -

`chgrp staff data.txt`

7. Why is it dangerous to give `777` permissions to a file? Explain with an example. (CO1)

Answer -

`777` means everyone can read, write and execute the file. Its risky because any user can modify or delete it.

Example -

`Chmod 777 important.txt`

8. What is the difference between `apropos` and `whatis`? (CO1)

Answer -

`apropos` - searches manual pages for a keyword.

`whatis` - shows a short description of a specific command.

9. Write a command to redirect the error output of a command to a file named `error.log`. (CO1)

Answer -

`command 2> error.log`

10. How can you use the `tee` command to append output to a file instead of overwriting it? (CO4)

Answer -

command | tee -a file.txt
(-a means append)